



BEST PRACTICE SERIES

Use of a “Hot Sheet” for Component Management

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

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1.0 Introduction

Condition Monitoring (CM) is the critical element to any successful repair-before-failure maintenance and repair (M&R) program. The only way to plan and execute repairs prior to failure is to have robust processes in place to detect and communicate the impending failures. Once the defects are detected there must be documentation and communication within the Planning department that allows them to adjust and define their short, medium and long-term M&R plans.

The **Hot Sheet** is a Condition Monitoring documentation and communication tool used specifically for **major components and systems**. Unscheduled component failures are often the costliest M&R events during the machine life cycle. They generally cause increased downtime, lost production, expediting costs, and additional collateral damage. The Hot Sheet is used to track the changes in component health in order to avoid these unexpected component failures.

2.0 Best Practice Description

2.1 Purpose:

In simplest terms the Hot Sheet is a tracking and communication tool for machine component status. It is a Condition Monitoring tool used by the Planning group to monitor component usage and watch for indications of developing failures. Typically spreadsheet based, it is a visual quick reference indicator of component health.

For a new machine, one of Planning's initial tasks is to develop the long-term M&R plan. By comparing historical component life data to planned machine usage, the Planner will estimate when the components will reach their useful life and therefore need replacement. This replacement schedule is the long-term plan. Usage is measured by fuel consumption for the engine and hour accumulation for all other components. As the machine matures, the Hot Sheet serves as the means to track and communicate the fuel consumption and hour accumulation within the Planning department. The Planner compares actual consumption and accumulation to the original estimated usage and makes the appropriate adjustments to the long-term plan. For example, if a particular machine is underutilized over time, then its component replacements can be delayed in the schedule.

Component health is also dependent on other Condition Monitoring routines. The Hot Sheet will also document Fluid Management results, (i.e. SOS, electronic machine data such as VIMS events). Any Condition Monitoring routine that indicates a change in component health can be included. *Overall the intent of the Hot Sheet is to capture and communicate to the Planner any indications of major component or system health changes that may require adjustments to the long-term plan.*

2.2 Owner:

Within the M&R Process model, the Fleet Analyst (FA) has responsibility for the definition of the Condition Monitoring routines and the subsequent data collection, analysis and technical interpretation. He is the on-site Condition Monitoring expert with the technical knowledge to translate the CM data into recommended M&R activities. The Fleet Analyst resides within the

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Planning group and works closely with the Planner to create action items based on Condition Monitoring results. Therefore it is the Fleet Analyst that will define and maintain the content of the Hot Sheet and collaborate with the Planner to define required adjustments to the long-term plan.

It is possible that a dealer structure consists of an on-site Fleet Analyst sharing responsibility with a central Fleet Analyst located at an off-site dealer office. The central FA may process VIMS and SOS results while the on-site FA handles all other Condition Monitoring data. Regardless, the on-site Fleet Analyst should own the Hot Sheet.

2.3 Data Sources, Content & Interpretations:

2.3.1 Fuel Consumption

Total fuel consumption during the period can be obtained from VIMS, the ECM or, on machines not equipped with VIMS, from fuel addition records.

Fuel consumption can be thought of as “engine life consumption”. By tracking actual fuel consumption versus total fuel consumed at engine rebuild, the Hot Sheet shows the percentage of engine life consumed to date (conversely remaining life). For machine ATY00744, Table 1 (page 4) shows that the engine has burned 161,760 gallons. 800,000 gallons are expected before a rebuild is required. Therefore the engine has used 20.22% of its expected life.

2.3.2 Machine Operating Hours

Operating hours are obtained from machine service meter readings. Note, hours obtained from dispatch systems frequently do not agree with machine SMU due to coding of production delays, etc. Note that hours taken from machine SMU will be higher than those taken from dispatch, often by as much as 10 percent.

Using a concept similar to fuel consumption to track utilized life for engines, all other components are tracked by their accumulated hours. In Table 1, the total accumulated hours are shown as a percentage of total expected hours before rebuild or replacement (benchmark hours).

If fuel or hours are accumulating at a different rate than the original estimate, the long-term plan for component rebuilds or replacements must be adjusted.

2.3.3 Fluids Management & Analyses

- SOS / particle count
- Kidney looping / fluid dialysis
- Magnetic plug inspections
- Filter / breather / strainer / screen inspections
- Misc. Fluid consumption ... oil, coolant, grease

Findings from Fluid Management CM routings may indicate that the component will not reach the original life defined by the long-term plan. For example, the transmission and torque converter SOS results for machine XYZ00744 show high particle count at only 56% of total expected life.

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The Fleet Analyst must determine if the component needs to be removed earlier than expected and therefore change Planning's medium or term plans.

2.3.4 On-board Electronic Data Monitoring

VIMS events of components operating out of recommended range or specification may indicate component deterioration.

Machine electronic data is used in the same manner as Fluid Management results to detect premature component deterioration. For machine XYZ00744, the VIMS events show transmission and torque converter slip times are higher than spec. This may indicate a faulty valve or worn clutch plates. In this case, the high particle count detected by SOS should also be considered during diagnosis.

2.3.5 Machine Systems Performance Tests

Pressure and cycle time tests may indicate component deterioration and should be documented and analyzed similar to VIMS data.

2.3.6 Application Analyses

Changes in application severity need to be reviewed periodically to determine any effect on component life expectations. For example, as the pit matures and deepens, the resulting increase in duty cycle severity on the drive train components may require reduced life estimates (benchmark hours).

2.3.7 Learning from Failures

Component failure analysis reports may yield failure modes unknown when the original life estimates were made and therefore require adjustments to expected life (benchmark hours).

Note: Areas of concern are color coded according to increasing risk. Refer to Table 1. Life accumulation of 75% and above is broken down into three escalating categories represented in yellow, red and black. Similarly Fluid Management and VIMS results are represented in the same manner of escalating risk.

793D Fleet Component Hot Sheet

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3/11/2008		Engine	Fuel Burn	Transmission	Torque	Differential	Rear	Final Drive
			Gallons				L	R
Serial Number	Benchmark Hours	18000	800000	14000	18000	14000	15000	15000
ATY00744	Current % of Benchmark	22.47%	20.22%	56.38%	53.22%	30.28%	28.26%	142.63%
164	Est Current Comp Hours	4044	161760	7893	9579	4239	4239	21395
	SOS			Hi particle ct. wear metals normal 2/20/08	Hi particle ct. wear metals normal 2/20/09			
500hr pm 3/11/08	VIMS			slip times hi >1 second	upshift lckup hi>1.45 secs			
ATY00745	Current % of Benchmark	24.94%	22.45%	24.04%	62.10%	24.04%	5.97%	142.63%
165	Est Current Comp Hours	4490	179600	3366	11178	3366	895	21395
	SOS							
	VIMS							
ATY00746	Current % of Benchmark	59.18%	53.27%	10.56%	8.21%	145.30%	55.28%	122.01%
166	Est Current Comp Hours	10653	426120	1478	1478	20342	8292	18301
	SOS							
	VIMS	low boost 22.7psi		slip times >1 second	upshift lckup hi>1.45 secs			
ATY00747	Current % of Benchmark	75.29%	67.76%	90.85%	70.66%	139.53%	43.76%	11.35%
167	Est Current Comp Hours	13552	542080	12719	12719	19534	6564	1702
	SOS							
1000hr pm 3/12/08	VIMS	overspeed @2332 rpm						
ATY00748	Current % of Benchmark	56.02%	50.42%	84.94%	66.07%	102.76%	105.83%	31.27%
168	Est Current Comp Hours	10084	403360	11892	11892	14386	15875	4690
	SOS							
	VIMS							
ATY00800	Current % of Benchmark	13.46%	12.12%	45.11%	56.16%	57.29%	113.89%	5.59%
169	Est Current Comp Hours	2423	96920	6316	10108	8021	17084	839
	SOS							
1000hr pm 3/11/08	VIMS							

Legend:

75% - 89% of LIFE
90% - 100% of LIFE
101% of LIFE and Greater

Table 1

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3.0 Implementation Steps

1. Create the Hot Sheet spreadsheet listing all on-site machines and their respective components that need to be tracked.
2. Leverage the Condition Monitoring infrastructure and processes to collect the Data Sources described above and in the Condition Monitoring BP **1007-2.0-1102**.
3. Assign responsibility within the Planning department to collect, analyze and populate the Hot Sheet with the Data Sources section 2.3 described above.
4. Define the communication channel or routing list for the Hot Sheet publication.
5. Establish the Hot Sheet master copy location and provide access to the members on the routing list. If possible, keep the master on a share drive with read/write access to the Fleet Analyst and read-only rights to everyone else.

4.0 Benefits

The on-time detection of potential problems is the key requisite to achieve the ultimate goal of "Repair Before Failure".

The communication of potential problems and deviations to expected hour accumulation allows Planning to:

1. Better coordinate upcoming component requirements with the Parts Department.
2. Schedule the necessary repairs at the right time (At the right window of opportunity and Before Failure)

4.1 Measurable Benefits

4.1.1 Time and Cost

The difference in downtime and cost between a repair executed before failure and one after failure has been established in the range of 5 to 8 times.

MTBS, a measure of machine health and repair quality, along with MTTR, a measure of repair efficiency, will both benefit from anticipated and planned component replacements that the Hot Sheet will facilitate. Otherwise, unexpected component failures are some of the most costly due to extensive machine downtime and potential collateral system damage and contamination.

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5.0 Resources Required

1. The most significant resource requirement will be having the Condition Monitoring systems, equipment and processes in place to provide the information described in the Data Sources section 2.3 and also covered in the Condition Monitoring BP **1007-2.0-1102**.
2. A Fleet Analyst or Planner (for small fleets) responsible for the Hot Sheet ownership.
3. Computer with spreadsheet capability and means to route the spreadsheet to the recipients either manually or electronically.

6.0 Supporting Attachments / References

[Component Hot Sheet example.xls](#)  (Click on "Attachments" tab within this document to view).

7.0 Related Best Practices

Condition Monitoring	BP 1007-2.0-1102
Planning & Scheduling	BP 1007-2.0-1104
Getting Started with MCTS	BP 0906-4.5-1013
MCTS used to Manage & Monitor Component Life	BP 0906-3.1-1020

8.0 Acknowledgements

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For further details and information regarding this Best Practice, please contact the author.

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